

Mixed-flavor neutrino trident production as a probe of lepton-flavor-violating Z' bosons at SBN, DUNE-ND, and FASER

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We propose neutrino trident production with a wrong-sign mixed-flavor final state, $\nu_\mu A \rightarrow \nu_e e^- \mu^+ A$, as a clean probe of lepton-flavor-violating Z' bosons with purely off-diagonal couplings in the e - μ sector. In the discrete-symmetry-protected limit, the new vector has no diagonal charged-lepton, quark, or photon-mixing couplings. As a result, the leading low-energy constraints arise from four-lepton processes such as muonium–antimuonium oscillation and from modifications of leptonic muon decay, rather than from tree-level $\mu \rightarrow 3e$ or one-loop $\mu \rightarrow e\gamma$. We compute the off-shell mixed-flavor trident matrix element in the Bethe–Heitler topology, including the unequal charged-lepton masses, and present its coherent EPA scaling. The wrong-sign topology is essentially absent in the Standard Model for a neutrino-mode beam, up to antineutrino beam contamination. The resulting search provides a complementary probe of the same dimension-six operator tested by muonium oscillations, with independent systematics and enhanced reach at high-energy neutrino facilities.

Introduction.—Lepton-flavor-violating (LFV) interactions of a neutral vector boson are usually severely constrained. If the vector also has flavor-diagonal charged-lepton or quark couplings, processes such as $\mu \rightarrow e\gamma$, $\mu \rightarrow 3e$, and coherent $\mu \rightarrow e$ conversion in nuclei impose strong limits on the underlying coupling. These constraints are greatly modified in the limit of *exact off-diagonality*, in which the new boson couples only between two different lepton flavors and diagonal couplings are forbidden by a discrete exchange symmetry. In such a scenario, the dominant probes are not necessarily the standard charged-lepton LFV searches, but rather four-lepton observables that directly involve two off-diagonal vertices.

In this Letter we consider the mixed-flavor trident process

$$\nu_\mu(p_1) A(P) \rightarrow \nu_e(p_2) e^-(k_-) \mu^+(k_+) A(P'), \quad (1)$$

and the charge-conjugate final state. The process is induced by a Z' exchanged between the neutrino line and the charged-lepton pair, with the coherent nuclear photon attaching to either of the outgoing charged leptons. Unlike same-flavor trident, the mixed-flavor final state directly tags the flavor-changing structure of the new current. For a neutrino-mode ν_μ beam, the $e^- \mu^+$ charge assignment is wrong-sign relative to the Standard Model charged-current trident process, which produces $\nu_e \mu^- e^+$ from an initial ν_μ . Consequently, the wrong-sign channel is a particularly clean search mode, limited primarily by antineutrino beam contamination and non-trident detector backgrounds.

We derive the corrected off-shell Z' matrix element and its leading coherent scaling. The key parametric result is simple. If the neutrino coupling is g_ν and the charged LFV couplings are g'_L and g'_R , then the coherent trident

cross section scales as

$$\sigma_{\text{coh}} \sim Z^2 \alpha^2 \frac{|g_\nu|^2 (|g'_L|^2 + |g'_R|^2)}{m_{Z'}^4} E_\nu^2 \times \text{logarithms and form-factor sup} \quad (2)$$

In the minimal symmetry-motivated model considered below, $g_\nu = g'_L$. For a purely left-handed charged current, $g'_R = 0$, the rate therefore scales as

$$\sigma_{\text{coh}} \propto Z^2 \alpha^2 \frac{|g'_L|^4}{m_{Z'}^4}, \quad (3)$$

not as a sixth power of the LFV coupling. Mixed trident, muonium–antimuonium oscillation, and leptonic muon decay therefore probe the same effective ratio $|g'_L|/m_{Z'}$ at leading order, but with different kinematics and systematics.

Model and existing constraints.—We consider a Z' boson with off-diagonal couplings between the first two lepton generations,

$$\mathcal{L}_{Z'} = g'_L (\bar{e} \gamma^\alpha P_L \mu + \bar{\nu}_e \gamma^\alpha P_L \nu_\mu) Z'_\alpha + g'_R \bar{e} \gamma^\alpha P_R \mu Z'_\alpha + \text{h.c.} \quad (4)$$

The symmetry limit of interest forbids diagonal $Z' \ell \bar{\ell}$ couplings, Z' couplings to quarks, and kinetic mixing with the photon. We keep g'_L and g'_R independent in the matrix-element formulae, while the minimal purely left-handed benchmark corresponds to $g'_R = 0$.

At momentum transfers small compared to the mediator mass, the interaction relevant for Eq. (1) is

$$\mathcal{L}_{\text{eff}} = -\frac{1}{m_{Z'}^2} [g_\nu \bar{\nu}_e \gamma^\alpha P_L \nu_\mu] [\bar{e} \gamma_\alpha (g'_L P_L + g'_R P_R) \mu] + \text{h.c.}, \quad (5)$$

with $g_\nu = g'_L$ for Eq. (4). This normalization keeps the chiral projectors explicit; alternative Fermi-like conventions may move factors of two or four into the definition of effective couplings.

Placeholder for Bethe–Heitler diagrams.

FIG. 1. Bethe–Heitler diagrams for $\nu_\mu A \rightarrow \nu_e e^- \mu^+ A$. The coherent photon emitted by the nucleus attaches to the outgoing electron or to the outgoing antimuon. Gauge invariance requires the sum of both attachments.

Muonium–antimuonium oscillation. Two insertions of the off-diagonal charged-lepton vertex generate four-lepton operators such as

$$\frac{g_L'^2}{m_{Z'}^2} (\bar{e} \gamma^\alpha P_L \mu) (\bar{e} \gamma_\alpha P_L \mu), \quad (6)$$

with analogous right-handed and mixed-chirality terms if $g_R' \neq 0$. Muonium–antimuonium oscillation therefore constrains the same coupling-over-mass combination that appears in the trident amplitude. For the left-handed benchmark, the bound may be expressed schematically as

$$|g_L'| \lesssim \text{few} \times 10^{-4} \left(\frac{m_{Z'}}{\text{GeV}} \right), \quad (7)$$

up to the precise normalization convention used for the four-fermion operator.

G_F universality. The same off-diagonal interaction contributes to $\mu^- \rightarrow e^- \nu_\mu \bar{\nu}_e$ through Z' exchange. This shifts the Fermi constant inferred from the muon lifetime relative to its extraction from semileptonic processes. The resulting constraint is again on $|g_L'|/m_{Z'}$ in the purely left-handed benchmark, with systematics independent from muonium oscillation.

Loop-induced charged-lepton flavor violation. In the exact off-diagonal limit, tree-level $\mu \rightarrow 3e$ requires a forbidden diagonal $\bar{e} e Z'$ vertex, and the usual one-loop dipole amplitude is absent in the symmetry limit. The leading nonzero dipole and three-body charged-lepton LFV amplitudes are therefore higher order in the symmetry-breaking effects or arise at higher loops, making them less constraining than the tree-level four-lepton probes in the parameter region considered here.

Trident cross section in the off-shell regime.—For mediator masses above the typical momentum exchange through the Z' line, the propagator may be expanded as

$$\frac{-i}{\ell^2 - m_{Z'}^2} \simeq \frac{i}{m_{Z'}^2}, \quad \ell = p_1 - p_2, \quad (8)$$

and the process is described by the contact interaction in Eq. (5). The coherent amplitude contains two powers of the electromagnetic coupling: one from the nuclear photon emission and one from its attachment to

the charged-lepton line. Consequently the coherent cross section is proportional to $Z^2 \alpha^2$, not to a single power of α .

The high-energy behavior is well captured by the leading-log estimate

$$\sigma_{\text{coh}} \sim Z^2 \alpha^2 \frac{|g_\nu|^2 (|g_L'|^2 + |g_R'|^2)}{m_{Z'}^4} E_\nu^2 \log^n \left(\frac{E_\nu^2}{m_{e\mu}^2} \right), \quad (9)$$

where n depends on the phase-space region retained by the coherent form factor, and $m_{e\mu}$ denotes the charged-lepton mass scale that regulates the collinear enhancement. For unequal final-state masses, the dominant logarithms are cut off by both m_e and m_μ , while the physical threshold is $m_e + m_\mu$.

Equation (9) is the scaling used for the qualitative discussion below. A numerical prediction should use the full Bethe–Heitler matrix element of the End Matter, the exact four-body phase space, and the nuclear form factor. The leading-log EPA formula is useful as an analytic guide, but not as a substitute for the full calculation in the threshold or moderate-energy regimes relevant to SBN and DUNE-ND.

Experimental opportunities.—The signal is a low-activity coherent event with two oppositely charged, different-flavor leptons and little or no hadronic recoil. Because the wrong-sign final state $e^- \mu^+$ is not produced by the dominant Standard Model charged-current trident process in a ν_μ beam, it provides a particularly clean signature of the off-diagonal interaction.

The relative strengths of SBN, DUNE-ND, and the LHC forward neutrino program follow from Eq. (9). SBN benefits from excellent imaging in liquid argon but operates near the mixed $e\mu$ threshold for sub-GeV neutrinos. DUNE-ND gains from higher energy, higher exposure, and detailed event reconstruction. FASER ν and the Forward Physics Facility gain from the much larger forward neutrino energies and high- Z tungsten or lead targets. The latter programs are therefore especially powerful in the off-shell regime, where the coherent cross section grows with neutrino energy until nuclear form-factor and phase-space effects regulate the enhancement.

A robust sensitivity projection should be based on the corrected matrix element below, not on the sixth-power scaling sometimes obtained by double-counting the charged LFV coupling. In the purely left-handed benchmark, fixed event yield corresponds approximately to a fixed value of $|g_L'|/m_{Z'}$, paralleling the scaling of muonium–antimuonium oscillation and G_F universality.

Backgrounds.—The leading irreducible Standard Model trident background in a neutrino-mode ν_μ beam is

$$\nu_\mu A \rightarrow \nu_e \mu^- e^+ A, \quad (10)$$

which populates the opposite charge assignment. The wrong-sign signal region $e^- \mu^+$ is therefore suppressed

Placeholder for corrected sensitivity figure.

FIG. 2. Projected sensitivity to the off-diagonal coupling as a function of $m_{Z'}$. Curves should be obtained with the corrected Bethe–Heitler matrix element and scale approximately with $|g'_L|/m_{Z'}$ in the purely left-handed off-shell benchmark.

by the antineutrino component of the beam. Reducible backgrounds include charged-current pion production with pion decay in flight, photon conversions, Dalitz decays, and heavy-flavor semileptonic decays at high energy. These processes are accompanied by displaced vertices, photon-conversion activity, or hadronic recoil, and can be controlled with vertex isolation, charge/flavor identification, and cuts on associated hadronic activity.

Discussion and conclusion.—Mixed-flavor neutrino trident production provides a direct probe of exactly off-

diagonal lepton currents. In the off-shell regime, the process is governed by the same dimension-six operator that controls muonium–antimuonium oscillation and leptonic muon decay, but it is tested in a high-energy scattering environment with different experimental systematics. The wrong-sign charge assignment in a neutrino beam removes the leading Standard Model trident background and makes this channel especially attractive for near detectors and forward LHC neutrino experiments.

The corrected matrix element shows that the purely left-handed rate scales as $|g'_L|^4/m_{Z'}^4$, not as a sixth power of the coupling. This is essential for interpreting the reach. Numerical projections should therefore be regenerated with the corrected Bethe–Heitler amplitude, including the proper antifermion propagator, coherent nuclear current, unequal charged-lepton masses, and exact phase-space boundaries.

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END MATTER

A. Corrected amplitude for mixed-flavor trident

We derive the matrix element for

$$\nu_\mu(p_1) + A(P) \rightarrow \nu_e(p_2) + e^-(k_-) + \mu^+(k_+) + A(P'), \quad (\text{A1})$$

in the coherent regime. The momentum transferred to the nucleus is

$$q = P - P', \quad (\text{A2})$$

and the momentum carried by the off-shell Z' is

$$\ell = p_1 - p_2. \quad (\text{A3})$$

Momentum conservation on the leptonic subdiagram gives

$$\ell + q = k_- + k_+. \quad (\text{A4})$$

The coherent nuclear electromagnetic current is written as

$$J_{\text{coh}}^\mu(P, P') = (P + P')^\mu F_A(q^2), \quad (\text{A5})$$

with the overall nuclear charge Ze displayed explicitly in the amplitude. In the heavy-target limit this reduces to $J_{\text{coh}}^\mu \simeq 2M_A \delta_0^\mu F_A(q^2)$.

The charged LFV vertex is

$$\Gamma^\alpha = \gamma^\alpha (g'_L P_L + g'_R P_R) = g'_V \gamma^\alpha - g'_A \gamma^\alpha \gamma_5, \quad (\text{A6})$$

where

$$g'_V = \frac{g'_L + g'_R}{2}, \quad g'_A = \frac{g'_L - g'_R}{2}. \quad (\text{A7})$$

For complex couplings, the conjugate vertex appearing in the squared amplitude is

$$\bar{\Gamma}^\beta \equiv \gamma^0 (\Gamma^\beta)^\dagger \gamma^0 = \gamma^\beta (g_L'^* P_L + g_R'^* P_R). \quad (\text{A8})$$

The Bethe–Heitler tensor for $\nu_\mu \gamma^* \rightarrow \nu_e e^- \mu^+$ is

$$\begin{aligned} T^{\mu\alpha} = e \bar{u}(k_-) & \left[\gamma^\mu \frac{\not{k}_- - \not{q} + m_e}{(k_- - q)^2 - m_e^2} \Gamma^\alpha \right. \\ & \left. + \Gamma^\alpha \frac{\not{q} - \not{k}_+ + m_\mu}{(q - k_+)^2 - m_\mu^2} \gamma^\mu \right] v(k_+). \end{aligned} \quad (\text{A9})$$

The second numerator is fixed by the fermion flow on the outgoing antimuon line. This sign is essential for the Ward identity.

The full amplitude in the off-shell limit $|\ell^2| \ll m_{Z'}^2$ is

$$\mathcal{M} = \frac{Ze}{q^2} J_\mu^{\text{coh}}(P, P') \frac{g_\nu}{m_{Z'}^2} [\bar{u}(p_2) \gamma_\alpha P_L u(p_1)] T^{\mu\alpha}, \quad (\text{A10})$$

where $g_\nu = g_L'$ in the model of Eq. (4). The overall phase is conventional and has no effect on the cross section.

Gauge invariance of the photon attachment requires both Bethe–Heitler diagrams. Contracting Eq. (A9) with q_μ and using the external equations of motion,

$$(\not{k}_- - m_e)u(k_-) = 0, \quad (\not{k}_+ + m_\mu)v(k_+) = 0, \quad (\text{A11})$$

one obtains

$$q_\mu T^{\mu\alpha} = 0. \quad (\text{A12})$$

This cancellation occurs between the electron and antimuon photon attachments and is lost if the antifermion propagator numerator is written with the wrong sign.

The charge-conjugate final state,

$$\nu_\mu A \rightarrow \nu_e e^+ \mu^-, \quad (\text{A13})$$

is obtained by replacing the charged-lepton spinors and using the Hermitian-conjugate charged LFV vertex. For real couplings and equal acceptances, the two charge configurations have equal rates in the absence of Standard Model interference.

B. Squared amplitude and coupling structure

Squaring Eq. (A10), summing over final spins, and not averaging over the initial neutrino helicity, gives

$$|\overline{\mathcal{M}}|^2 = \frac{Z^2 e^2}{q^4} \frac{|g_\nu|^2}{m_{Z'}^4} J_\mu^{\text{coh}} J_\nu^{\text{coh}*} L_{\alpha\beta} W^{\mu\alpha, \nu\beta}. \quad (\text{A14})$$

The neutrino tensor, with the coupling g_ν factored outside, is

$$\begin{aligned} L_{\alpha\beta} &= \text{Tr} \left[\not{p}_2 \gamma_\alpha P_L \not{p}_1 \gamma_\beta P_L \right] \\ &= 2 [p_{1\alpha} p_{2\beta} + p_{1\beta} p_{2\alpha} - g_{\alpha\beta} (p_1 \cdot p_2) - i \epsilon_{\alpha\beta\rho\sigma} p_1^\rho p_2^\sigma], \end{aligned} \quad (\text{A15})$$

where the sign of the antisymmetric term follows the convention $\epsilon^{0123} = +1$ and the mostly-minus metric. If instead one uses currents normalized as $\gamma^\mu(1 - \gamma_5)$ rather than $\gamma^\mu P_L$, this tensor is larger by a factor of four; the coupling normalization must then be changed consistently.

The charged Bethe–Heitler tensor is

$$W^{\mu\alpha, \nu\beta} = \sum_{\text{spins}} T^{\mu\alpha} T^{\nu\beta*}. \quad (\text{A16})$$

It can be decomposed into electron-attachment, antimuon-attachment, and interference terms. Defining

$$D_e = (k_- - q)^2 - m_e^2, \quad D_\mu = (q - k_+)^2 - m_\mu^2, \quad (\text{A17})$$

one obtains

$$W_{ee}^{\mu\alpha, \nu\beta} = \frac{e^2}{D_e^2} \text{Tr} \left[(\not{k}_- + m_e) \gamma^\mu (\not{k}_- - \not{q} + m_e) \Gamma^\alpha (\not{k}_+ - m_\mu) \bar{\Gamma}^\beta \right. \\ \left. \times (\not{k}_- - \not{q} + m_e) \gamma^\nu \right], \quad (\text{A18})$$

$$W_{\mu\mu}^{\mu\alpha, \nu\beta} = \frac{e^2}{D_\mu^2} \text{Tr} \left[(\not{k}_- + m_e) \Gamma^\alpha (\not{q} - \not{k}_+ + m_\mu) \gamma^\mu (\not{k}_+ - m_\mu) \right. \\ \left. \times \gamma^\nu (\not{q} - \not{k}_+ + m_\mu) \bar{\Gamma}^\beta \right], \quad (\text{A19})$$

and

$$W_{e\mu}^{\mu\alpha, \nu\beta} = \frac{e^2}{D_e D_\mu} \text{Tr} \left[(\not{k}_- + m_e) \gamma^\mu (\not{k}_- - \not{q} + m_e) \Gamma^\alpha (\not{k}_+ - m_\mu) \right. \\ \left. \times \gamma^\nu (\not{q} - \not{k}_+ + m_\mu) \bar{\Gamma}^\beta \right], \quad (\text{A20})$$

with

$$W_{\mu e}^{\mu\alpha, \nu\beta} = (W_{e\mu}^{\nu\beta, \mu\alpha})^*. \quad (\text{A21})$$

The full tensor is

$$W^{\mu\alpha, \nu\beta} = W_{ee}^{\mu\alpha, \nu\beta} + W_{\mu\mu}^{\mu\alpha, \nu\beta} + W_{e\mu}^{\mu\alpha, \nu\beta} + W_{\mu e}^{\mu\alpha, \nu\beta}. \quad (\text{A22})$$

The coupling dependence follows directly from Eqs. (A6) –(A8). In the limit where chirality-flipping terms are neglected, the squared amplitude is proportional to

$$|g_\nu|^2 (|g'_L|^2 + |g'_R|^2). \quad (\text{A23})$$

Terms proportional to $\text{Re}(g'_L g'^*_R)$ require a chirality flip on the charged-lepton line and are therefore accompanied by powers of $m_e m_\mu$ in the trace. Thus a useful schematic form is

$$\overline{|\mathcal{M}|^2} = \frac{Z^2 e^4 F_A^2(q^2)}{q^4 m_{Z'}^4} |g_\nu|^2 \left[(|g'_L|^2 + |g'_R|^2) S_1 \right. \\ \left. + 2 \text{Re}(g'_L g'^*_R) m_e m_\mu S_2 \right], \quad (\text{A24})$$

where S_1 and S_2 are real functions of the independent Lorentz invariants. In the model of Eq. (4), $g_\nu = g'_L$. For the purely left-handed benchmark, $g'_R = 0$, one obtains

$$\overline{|\mathcal{M}|^2} \propto \frac{Z^2 e^4 F_A^2(q^2)}{q^4 m_{Z'}^4} |g'_L|^4. \quad (\text{A25})$$

This fourth-power scaling in g'_L is the appropriate off-shell trident scaling. A sixth-power dependence would double-count one of the charged-current LFV vertices.

C. EPA scaling and phase-space representation

The exact coherent cross section is

$$d\sigma = \frac{1}{4(P \cdot p_1)} \overline{|\mathcal{M}|^2} (2\pi)^4 \delta^{(4)}(p_1 + P - p_2 - k_- - k_+ - P') \\ \times \frac{d^3 p_2}{(2\pi)^3 2E_2} \frac{d^3 k_-}{(2\pi)^3 2E_-} \frac{d^3 k_+}{(2\pi)^3 2E_+} \frac{d^3 P'}{(2\pi)^3 2E_{P'}}. \quad (\text{A26})$$

This is the expression to be used for numerical work.

In the high-energy coherent regime, one may use an equivalent-photon approximation to organize the leading logarithms. Schematically,

$$\sigma_{\text{coh}} = \int dQ^2 d\hat{s} P_A(Q^2, \hat{s}) \sigma(\nu_\mu \gamma \rightarrow \nu_e e^- \mu^+; \hat{s}), \quad (\text{A27})$$

where $Q^2 = -q^2$, $\hat{s} = (p_1 + q)^2$ is the invariant mass squared of the neutrino-photon subprocess, and P_A is the coherent nuclear photon flux including the exact kinematic limits and the factor $Z^2 F_A^2(Q^2)$. The EPA is controlled by small photon virtuality and by the nuclear coherence condition $Q \lesssim R_A^{-1}$. Coherence does not require $Q^2 \ll m_e^2$; for medium and heavy nuclei one often has $Q^2 \gg m_e^2$ while still remaining in the coherent regime.

The leading high-energy scaling of the subprocess is

$$\sigma(\nu_\mu \gamma \rightarrow \nu_e e^- \mu^+; \hat{s}) \sim \alpha \frac{|g_\nu|^2 (|g'_L|^2 + |g'_R|^2)}{m_{Z'}^4} \hat{s} \log^2 \left(\frac{\hat{s}}{m_{e\mu}^2} \right), \quad (\text{A28})$$

up to order-one constants, nonlogarithmic terms, and unequal-mass threshold corrections. The mass scale $m_{e\mu}$ is set by the charged-lepton masses and should not be replaced by a single universal mass away from the strict leading-log limit. The exact unequal-mass subprocess is obtained by integrating Eq. (A14) over the three-body final-state phase space.

Combining the EPA photon flux with Eq. (A28) gives the parametric coherent scaling

$$\sigma_{\text{coh}} \sim Z^2 \alpha^2 \frac{|g_\nu|^2 (|g'_L|^2 + |g'_R|^2)}{m_{Z'}^4} E_\nu^2 \times \text{logarithms and nuclear-form-factor suppression.} \quad (\text{A29})$$

For $g_\nu = g'_L$ and $g'_R = 0$, this reduces to

$$\sigma_{\text{coh}} \propto Z^2 \alpha^2 \frac{|g'_L|^4}{m_{Z'}^4}. \quad (\text{A30})$$

Equations (A29) and (A30) are the analytic scaling relations. A fully differential expression in variables such as Q^2 , $\hat{s}$, the charged-lepton energies, and the azimuthal angles should be obtained from the exact phase-space formula rather than from a simplified $dE_\gamma dQ^2$ photon flux.

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